

Local optima

OVA · 2026 · Andrew Lamont

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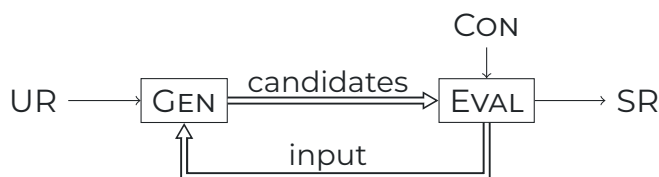
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[Harmonic Serialism](#)¹ (Prince and Smolensky, 1993/2004; McCarthy, 2000) – or more aptly *persistent OT* (McCarthy, 2006, 2007)

¹ [Harmonic Serialism](#)

- CON: ranked phonotactic and faithfulness constraints
no [Correspondence Theory](#)² (McCarthy and Prince, 1994, 1995, 1999)
- GEN: returns candidates that differ from input with at most one operation application
requires explicit hypotheses about operations
- Generate and evaluate until input is optimal

² [Correspondence Theory](#)



Today: evidence for HS from non-optimal mappings

1 Nasal prefixes in Bukusu ([bxk](#), Niger-Congo)

Three processes at prefix-root boundary; let's ignore the behavior of the infinitive and the tonal alternations

- Post-nasal voicing
/N-[t](#)ím-a/ → [n-[d](#)ím-a]
- Post-nasal hardening
/N-[β](#)ukul-a/ → [m-[b](#)ukúl-a]
- Nasal deletion
/N-[sáβ](#)-a/ → [sáβ-a]

(1) **Segmental alternations in Bukusu** (Mutonyi, 2000:168, 173-175)

	/xu-xu-/	/N-/	
a. /-tim-/	xu:-tim-a	n-dím-a	'to run / I run'
b. /-kaβ-/	xu:-kaβ-a	ŋ-gáβ-a	'to divide / I divide'
c. /-βukul-/	xú:-βukúl-á	m-bukúl-a	'to take / I take'
d. /-rafu-/	xú:-rafú-á	n-dafú-a	'to get angry / I get angry'
e. /-xalakil-/	xú:-xalákíl-á	xalákil-a	'to cut fast; to decide / I cut fast; I decide'
f. /-saβ-/	xu:-saβ-a	sáβ-a	'to beg / I beg'

1.1 Parallel OT does not work

These alternations don't work in OT with these constraints

- $*[+nasal][-voice]$: Assign One Violation For Every nasal followed by a voiceless obstruent.³
- $*[+nasal][+cont]$: AOVFE nasal followed by a continuant obstruent.⁴
- IDENT(nasal): AOVFE changed [nasal] feature.⁵
- IDENT(voice): AOVFE changed [voice] feature.
- IDENT(cont): AOVFE changed [continuant] feature.
- DEP: AOVFE inserted segment.⁶
- MAX: AOVFE deleted segment.⁷
- MAX/root: AOVFE deleted root segment.

³ Pater (1999:313)

⁴ Zsiga et al. (2006:725)

⁵ McCarthy and Prince (1994:340)

⁶ McCarthy and Prince (1994:342)

⁷ McCarthy and Prince (1994:116, 342)

Changing [voice] (e) is preferred to changing [nasal] (b), epenthesis (c), and deletion (d)⁸

(2) /N-tím-a/ → [n-dím-a]

/N-tím-a/	$*[+nasal][-voice]$	$*[+nasal][+cont]$	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [ntíma]	W 1					L		
b. [dtíma]			W 1			L		
c. [nətíma]				W 1		L		
d. [tíma]					W 1	L		
 e. [ndíma]						1		

⁸ W = more violations than winner; L = fewer violations than winner; rankings are valid when every L is dominated by some W (Prince, 2002)

Changing [continuant] (e) is preferred to changing [nasal] (b), epenthesis (c), and deletion (d)

(3) /N-βukul-a/ → [m-bukúl-a]

/N-βukul-a/	*[+nasal][-voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [mβukúla]		W 1					L	
b. [bβukúla]			W 1				L	
c. [nəβukúla]				W 1			L	
d. [βukúla]					W 1		L	
 e. [mbukúla]							1	

Deletion (d) is preferred to changing [voice] and [cont] (f)

(4) /N-sáβ-a/ → [sáβ-a]

/N-sáβ-a/	*[+nasal][-voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [nsáβa]	W 1	W 1			L			
b. [dsáβa]			W 1		L			
c. [nəsáβa]				W 1	L			
 d. [sáβa]					1			
e. [náβa]					1			W 1
 f. [ndáβa]					L	W 1	W 1	

Parallel OT does not predict this

- Deletion is more expensive than changing features

$$\langle 0, 0, 0, 0, 1, 0, 0, 0 \rangle > \langle 0, 0, 0, 0, 0, m, n, 0 \rangle \quad \forall m, n$$

- Changing both features is the global optimum;⁹ its violation vector¹⁰ is the lowest

$$\langle 0, 0, 0, 0, 0, 1, 1, 0 \rangle < \langle 0, 0, 0, 0, 1, 0, 0, 0 \rangle$$

- OT only returns global optima

“for a given input i, a grammatical structural description or output A is one that has the

⁹ global optimum

¹⁰ violation vector

greatest Harmony among the set $\text{Gen}(i)$ of all possible parses of i . That is, a parse A of i is grammatical if and only if there is no competing parse B in $\text{Gen}(i)$ such that B has higher Harmony than A " (Prince and Smolensky, 1993/2004:82)

Harmony $\sim$ -violation vector

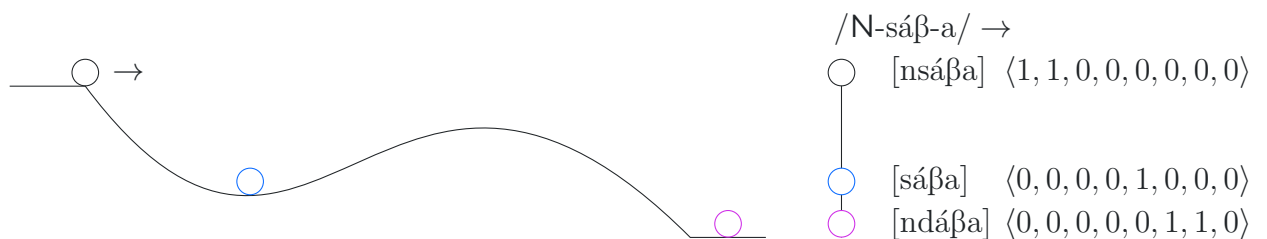
Not all attested mappings are global optima

- For a given constraint set
- Can introduce **conjoined constraints**¹¹ (Smolensky, 1993) or **weighted constraints**¹² (Farris-Trimble, 2008)
- But some mappings are instead **local optima**¹³
- McCarthy's (2007) potential energy analogy

¹¹ **conjoined constraints**

¹² **weighted constraints**

¹³ **local optima**



- If the grammar has to explore the candidate space step-by-step, it can get stuck in a local optimum

1.2 Harmonic Serialism works

Operations for Bukusu¹⁴

Feature changes $[\alpha F] \rightarrow [-\alpha F]$
 Deletion $/\alpha/ \rightarrow [\epsilon]$
 Epenthesis $/\epsilon/ \rightarrow [\alpha]$

Constraints same as before

¹⁴ For discussion of deletion as an operation see McCarthy (2008b, 2019) and Lin (2014) for epenthesis

(5) /N-tím-a/ → [n-dím-a]

/N-tím-a/	*[+nasal][−voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [Ntíma]	W 1					L		
b. [ʔtíma]			W 1			L		
c. [Nətíma]				W 1		L		
d. [tíma]					W 1	L		
☞ e. [Ndíma]						1		
f. [Ndíma]								
☞ g. [ndíma]								
☞ h. [ndíma]								

- [ndíma] is not a candidate in the first step
GEN cannot change [place] and [voice] simultaneously more on the second step in a future class
- Faithfulness constraints relative to input not UR
- Derivation **converges**¹⁵ once input is optimal

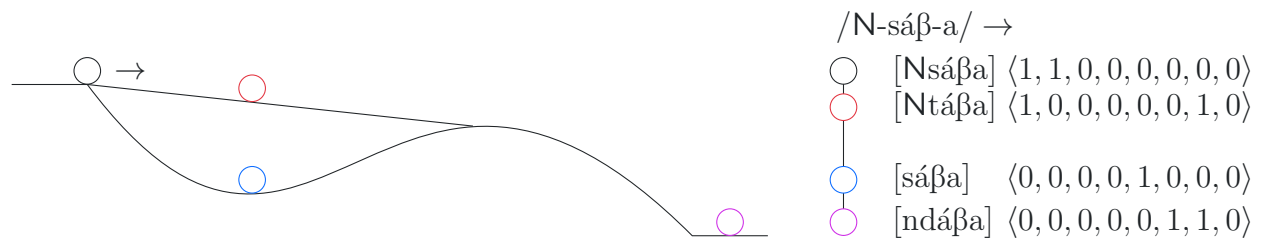
¹⁵ **converges**

(6) /N-sáβ-a/ → [sáβ-a]

/N-sáβ-a/	*[+nasal][−voice]	*[+nasal][+cont]	IDENT(nasal)	DEP	MAX	IDENT(voice)	IDENT(cont)	MAX/root
a. [Nsáβa]	W 1	W 1			L			
b. [ʔsáβa]			W 1		L			
c. [Nəsáβa]				W 1	L			
☞ d. [sáβa]					1			
e. [Náβa]					1			W 1
f. [Ntáβa]	W 1				L		W 1	
g. [Nzáβa]		W 1			L	W 1		
☞ h. [sáβa]								

- Deletion satisfies both phonotactic constraints
- [Ndáβa] is not a candidate in the first step

cannot change [continuant] and [voice] simultaneously
 every step picks optimal candidate; cannot reach [ndáβa]



- [sáβa] is local optimum in step 1, not global optimum
- HS is greedy: every step improves as much as possible

2 High tone in Jita (jit, Niger-Congo)

Tones shift one syllable to the right

- let's ignore gliding and vowel coalescence
- the data are more complicated; this is a simplification

(7) Tone shift in Jita (Downing, 2014:103-106)

- | | | | |
|----|--------------------|--------------------|----------------------|
| a. | /oku-βum-an-a/ | [oku-βum-an-a] | 'to hit each other' |
| b. | /oku-βón-an-a/ | [oku-βon-án-a] | 'to see each other' |
| c. | /oku-ljá munó/ | [oku-lja múnó] | 'to eat a lot' |
| d. | /oku-βón-a i:ɲɔɲi/ | [oku-βon-á i:ɲɔɲi] | 'to see a bird' |
| e. | /oku-mú-βum-ir-a/ | [oku-mu-βúm-ir-a] | 'to hit for him/her' |
| f. | /oku-mú-βon-a/ | [oku-mu-βón-a] | 'to see him/her' |
| g. | /a-amá-gur-a/ | [a-ama-gúr-a] | 's/he bought' |

- Bickmore (1999:130) posits a phonotactic constraint that requires high tone spans to contain exactly two TBUs, analogizing to FTBIN (see also McCollum and Kavitskaya, 2022)

Breteler (2017, 2018) derives binarity as footing

VVVV → V(VV)V → V(VV)V (→ V(VV)V)

2.1 Harmonic Serialism works

Operations for Jita¹⁶

- Insert an association line (without creating a gap)
- Delete an association line (without creating a gap)
- Build a foot

Constraints

- LICENSE(H, F):¹⁷ AOVFE H not linked to footed syllable
- LICENSE(F, H): AOVFE foot w/o a H-toned syllable
- *H:¹⁸ AOVFE high-toned syllable
- ALL-FEET-RIGHT:¹⁹ For every foot, assign as many violations as there are syllables to its right
- ALIGN-R(H, F): AOVFE H not right-aligned to a foot
- DEP(link):²⁰ AOVFE inserted association line
- MAX(link): AOVFE deleted association line

¹⁶ For discussion of autosegmental operations see Kimper (2012), McCarthy et al. (2012), Gietz et al. (2023), and Lamont (2025) and for footing see Pruitt (2010, 2012)

¹⁷ Breteler (2018:19) after Zoll (1996)

¹⁸ Yip (2002:80)

¹⁹ McCarthy and Prince (1993:80)

²⁰ Yip (2002:79)

(8) /oku-mú-βon-ir-a/ → [oku-mu-βón-ir-a], Step 1

okumuβonira H	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	*H	MAX(link)	LICENSE(F, H)
a. okumuβonira H	W 1	L	1		1		
b. okumuβonira H	W 1	L	1	W 1	W 2		
c. okumuβonira H	W 1	L	1		L	W 1	
d. o(kumu)βonira H		W 3	L		1		
e. oku(muβo)nira H		2	1		1		
f. okumuβo(nira) H	W 1	L	1		1		W 1

(9) /oku-mú-βon-a/ → [oku-mu-βón-a], Steps 2-4

	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	*H	MAX(link)	LICENSE(F, H)
H oku(muβo)nira							
g. H oku(muβo)nira		2	W 1	L	L 1		
h. H oku(muβo)nira		2		1	2		
i. H oku(muβo)(nira)		2	W 1	L	L 1		W 1
j. H oku(muβo)nira		2			W 2	L	
k. H oku(muβo)nira		2		W 1	W 3		
l. H oku(muβo)(nira)		2			W 2		W 1
m. H oku(muβo)nira		2			1	1	
n. H oku(muβo)nira		2	W 1		1	1	
o. H oku(muβo)nira		2			1		

The SR is not the global optimum

(10) /oku-mú-βon-a/ → [oku-mu-βón-a] in parallel OT

H okumuβonira	LICENSE(H, F)	ALL-FEET-R	ALIGN-R(H, F)	DEP(link)	*H	MAX(link)	LICENSE(F, H)
a. H okumuβonira	W 1	L	W 1	L	1	L	
☹ b. H oku(muβo)nira		2		1	1	1	
💣* c. H okumuβo(nira)		L		1	1	1	

3 Local optionality

Variation as randomly varying ranking (Kimper, 2011)
French ([fra](#), Indo-European)

(11) **Local optionality in French** (Kimper, 2011:424)

envie de te le demander

'feel like asking you for it'

[ãvi də tə lə dəmə̃də]

[ãvi d_ tə lə dəmə̃də]

[ãvi də t_ lə dəmə̃də]

[ãvi də tə l_ dəmə̃də]

[ãvi də tə lə d_ m̃də]

[ãvi d_ tə l_ dəmə̃də]

[ãvi də t_ lə d_ m̃də]

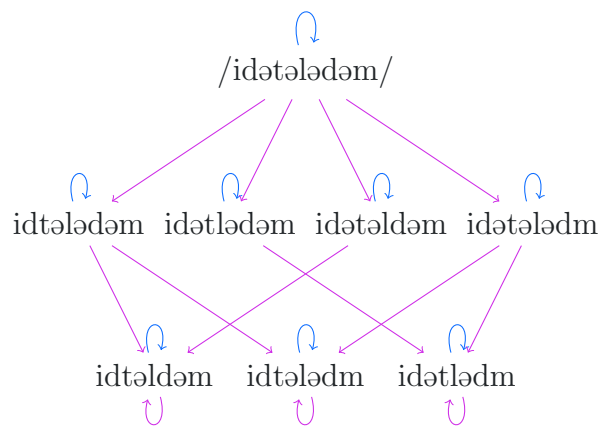
A gross oversimplification

- *CCC: AOVFE triconsonantal cluster
- *_ə
- MAX

Operation: /ə/ → ∅ (McCarthy, 2019)

*CCC ≫ {*_ə, MAX}

(12) **All possible derivations;** MAX $\gg$ *_ə vs. *_ə $\gg$ MAX



(13)

/...idətələdəm.../	*CCC	* _ə	MAX
a. [...idətələdəm...]		4	
b. [...idtələdəm...]		3	1
c. [...idətlədəm...]		3	1
d. [...idətəldəm...]		3	1
e. [...idətələdm...]		3	1

Not possible in parallel OT (see Riggle and Wilson, 2005 and Kaplan, 2016 for discussion)

(14)

/...idətələdəm.../	*CCC	* _ə	MAX
a. [...idətələdəm...]		4	
☞ b. [...idtələdəm...]		3	1
☞ c. [...idətlədəm...]		3	1
☞ d. [...idətəldəm...]		3	1
☞ e. [...idətələdm...]		3	1
f. [...idtəldəm...]		2	2
g. [...idətlədm...]		2	2
h. [...idtəldəm...]		2	2

okay, but free choice is not a good model of variation

- EVAL cannot distinguish the tied candidates
- impossible to build in probabilistic preferences (McCarthy, 2008a:§6.2)

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